

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dot Product, Projections & Orthogonality	26 PYQs	Scalar Product
Very High	Cross Product & Geometric Areas	16 PYQs	Vector Product
Moderate	Scalar Triple Product (Box Product) & Coplanarity	5 PYQs	Volume & Coplanarity
Standard	Vector Triple Product & Lagrange Identity	1 PYQs	Triple Products
Standard	Linear Combination & Vector Equations	1 PYQs	Vector Equations
Standard	Section Formula, Collinearity & Centroids	0 PYQs	Geometry & Position Vectors

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dot Product, Projections & Orthogonality	$\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos\theta$, $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$, $\text{Proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{ \vec{b} ^2} \vec{b}$
Cross Product & Geometric Areas	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$, $\text{Area}_{\triangle} = \frac{1}{2} \vec{a} \times \vec{b} $, $\hat{n} = \frac{\vec{a} \times \vec{b}}{ \vec{a} \times \vec{b} }$
Scalar Triple Product (Box Product) & Coplanarity	$[\vec{a}; \vec{b}; \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$, $[\vec{a}; \vec{b}; \vec{c}] = 0 \iff \text{coplanar}$
Vector Triple Product & Lagrange Identity	$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$, $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
Linear Combination & Vector Equations	$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, $\text{or take } \vec{a} \times (\vec{r} \times \vec{b}) \text{ to decouple}$
Section Formula, Collinearity & Centroids	$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$, $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$, $\vec{a} = \lambda \vec{b} \iff \text{collinear}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dot Product, Projections & Orthogonality (Very High Yield — 26 PYQs)

Core Principle: Scalar product $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_x b_x + a_y b_y + a_z b_z$. Orthogonality condition: $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$. Scalar projection of $\vec{a}$ on $\vec{b}$: $\text{Proj}_{\vec{b}}(\vec{a}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$.

■ **Caution / JEE Pitfall:** 1. Vector projection of $\vec{a}$ along $\vec{b}$ is $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\right)\vec{b}$. 2. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b})$ (do not write $2|\vec{a}||\vec{b}|$ unless collinear!).

■ **Speed Trick:** Unit vector along the angle bisector of $\vec{a}$ and $\vec{b}$: $\vec{u} = \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a}=2\hat{i}+\lambda\hat{j}+k\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+3k\hat{k}$ are orthogona..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\vec{a}=\hat{i}+\hat{j}+k\hat{k}, \vec{b}=\hat{j}-k\hat{k}$ and $\vec{a} \times \vec{c}=\vec{b}, \vec{a} \cdot \vec{c}=3$, then $\vec{c}$ i..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $\vec{a}=2\hat{i}+2\hat{j}-k\hat{k}, \vec{b}=\alpha\hat{i}+\beta\hat{j}+2k\hat{k}$ and $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$, then $\alpha+\beta=...$ " [Ans: D]

3.2 Cross Product & Geometric Areas (Very High Yield — 16 PYQs)

Core Principle: Vector cross product $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \hat{n}$. Collinearity condition: $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$. Area of triangle with sides $\vec{a}, \vec{b}$ is $(1/2)|\vec{a} \times \vec{b}|$; parallelogram area is $|\vec{a} \times \vec{b}|$; parallelogram with diagonals $\vec{d}_1, \vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$.

■ **Caution / JEE Pitfall:** 1. Cross product is anti-commutative: $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. 2. Lagrange's identity: $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.

■ **Speed Trick:** Area of a parallelogram given diagonal vectors $\vec{d}_1, \vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$, NOT $|\vec{d}_1 \times \vec{d}_2|$!

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If $\vec{a}=\hat{i}+\hat{j}+k\hat{k}, \vec{b}=\hat{j}-k\hat{k}$ and $\vec{a} \times \vec{c}=\vec{b}, \vec{a} \cdot \vec{c}=3$, then $\vec{c}$ i..." [Ans: C]
- [2025 Shift] Q#1 (MCQ): "If $|\vec{a}|=10, |\vec{b}|=2$ and $\vec{a} \cdot \vec{b}=12$, then the value of $|\vec{a} \times \vec{b}|$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Consider the following statements : Statement (I) : If either $|\vec{a}|=0$ or $|\vec{b}|=0$, then $\vec{a} \cdot \vec{b}=0$ Statement (II)..." [Ans: N/A]

3.3 Scalar Triple Product (Box Product) & Coplanarity (Moderate Yield — 5 PYQs)

Core Principle: Scalar triple product $[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \det([\vec{a} \ \vec{b} \ \vec{c}])$. Three vectors are coplanar if and only if $[\vec{a} \ \vec{b} \ \vec{c}] = 0$. Volume of parallelepiped $V = |[\vec{a} \ \vec{b} \ \vec{c}]|$; tetrahedron volume $V_{\text{tet}} = (1/6)|[\vec{a} \ \vec{b} \ \vec{c}]|$.

■ **Caution / JEE Pitfall:** 1. Cyclic order preserves sign: $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$. Non-cyclic swap negates: $[\vec{a} \ \vec{c} \ \vec{b}] = -[\vec{a} \ \vec{b} \ \vec{c}]$. 2. If any two vectors are identical or collinear, the box product is strictly 0.

■ **Speed Trick:** Four position points A, B, C, D are coplanar if and only if $[\vec{AB} \ \vec{AC} \ \vec{AD}] = 0$.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors and $\vec{p}, \vec{q}$ and $\vec{r}$ are vectors defined by $\vec{p} = \frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$, $\vec{q} = \frac{1}{3}(\vec{a} + \vec{b} - \vec{c})$, $\vec{r} = \frac{1}{3}(\vec{a} - \vec{b} + \vec{c})$ [Ans: N/A]
- [2020 Shift] Q#1 (MCQ): "If the vectors $2\vec{i} + 3\vec{j} + 4\vec{k}$, $2\vec{i} + \vec{j} - \vec{k}$ and $\lambda\vec{i} - \vec{j} + 2\vec{k}$ are coplanar, then the value of λ is..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "The two vectors $\vec{i} + \vec{j} + \vec{k}$ and $\vec{i} + 3\vec{j} + 5\vec{k}$ represent the two sides $\overline{AB}$ and $\overline{AC}$ respectively of a ΔABC . The length of the median through A is..." [Ans: N/A]

3.4 Vector Triple Product & Lagrange Identity (Standard Yield — 1 PYQs)

Core Principle: Vector triple product expands via the BAC-CAB identity: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$. It lies in the plane of $\vec{b}$ and $\vec{c}$, perpendicular to $\vec{a}$.

■ **Caution / JEE Pitfall:** Vector triple product is non-associative: $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$. Notice that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$.

■ **Speed Trick:** Mnemonic: 'BAC minus CAB' — dot product of outer vectors multiplied by inner vector, minus dot product of first two multiplied by the third.

Representative Source PYQs:

- [2024 Shift] Q#3 (MCQ): "The vectors $\vec{AB} = 3\vec{i} + 4\vec{k}$ and $\vec{AC} = 5\vec{i} - 2\vec{j} + 4\vec{k}$ are the sides of a ΔABC . The length of the median through A is..." [Ans: N/A]

3.5 Linear Combination & Vector Equations (Standard Yield — 1 PYQs)

Core Principle: Solving vector equations like $\vec{r} \times \vec{a} = \vec{b}$ and $\vec{r} \cdot \vec{c} = d$. Express $\vec{r}$ as a linear basis expansion $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$ or take vector cross/dot products with known vectors on both sides.

■ **Caution / JEE Pitfall:** When $\vec{r} \times \vec{a} = \vec{b}$, note that $\vec{b} \perp \vec{a}$ (so $\vec{a} \cdot \vec{b} = 0$). The general solution is $\vec{r} = \lambda \vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$.

■ **Speed Trick:** If $\vec{r} \times \vec{a} = \vec{b}$, taking the cross product with $\vec{a}$ gives $\vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$, decoupling $\vec{r}$ directly.

Representative Source PYQs:

- [2024 Shift] Q#4 (MCQ): "The volume of the parallelepiped whose co terminous edges are $\vec{j} + \vec{k}$, $\vec{i} + \vec{k}$ and $\vec{i} + \vec{j}$ is..." [Ans: N/A]

3.6 Section Formula, Collinearity & Centroids (Standard Yield — 0 PYQs)

Core Principle: Section formula for point dividing line segment in ratio $m:n$: $\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$ (internal) and $\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}$ (external). Centroid of triangle $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$. Three points are collinear if $\vec{AB} = \lambda \vec{BC}$.

■ **Caution / JEE Pitfall:** Collinear vectors can be antiparallel (negative scalar λ). External division formula carries an explicit minus sign in numerator and denominator: $\frac{m\vec{b} - n\vec{a}}{m-n}$.

■ **Speed Trick:** In a tetrahedron with vertices $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, the centroid is $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + k\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3k\hat{k}$ are orthogona..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\vec{a} = \hat{i} + \hat{j} + k\hat{k}, \vec{b} = \hat{j} - k\hat{k}$ and $\vec{a} \times \vec{c} = \vec{b}, \vec{a} \cdot \vec{c} = 3$, then $\vec{c}$ i..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value of λ for which the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + k\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3k\hat{k}$ are orthogonal is

- (A) $(5/2)$
- (B) $(-5/2)$ [Correct]
- (C) $(2/5)$
- (D) $(-2/5)$

Correct Answer: B

Solution Analysis: For two vectors to be orthogonal, their dot product must be zero. Given, $\vec{a} = 2\hat{i} + \lambda\hat{j} + k\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3k\hat{k}$. Now, $\vec{a} \cdot \vec{b} = 2(1) + \lambda(2) + 1(3) = 2 + 2\lambda + 3 = 5 + 2\lambda$. Since the vectors are orthogonal, $5 + 2\lambda = 0$
 $2\lambda = -5 \Rightarrow \lambda = (-5/2)$ So, the correct option is Option B $(-5/2)$

Benchmark Problem 2: KCET 2024 (Q#3)

The vectors $\vec{AB} = 3\hat{i} + 4k\hat{k}$ and $\vec{AC} = 5\hat{i} - 2\hat{j} + 4k\hat{k}$ are the sides of a $\triangle ABC$, The length of the median through A is

- (A) $\sqrt{18}$
- (B) $\sqrt{72}$
- (C) $\sqrt{33}$
- (D) $\sqrt{288}$

Correct Answer: None

Solution Analysis: Let A = origin $\therefore$ $\vec{AB}$ = Position vector of B $\vec{AC}$ = Position vector of C $\therefore$ Position vector of mid-point of B and C $= \frac{\vec{AB} + \vec{AC}}{2} = \frac{(3+5)/2)\hat{i} + ((0-2)/2)\hat{j} + ((4+4)/2)\hat{k}}{2} = 4\hat{i} - \hat{j} + 4k\hat{k}$ Median = $\vec{AP} = 4\hat{i} - \hat{j} + 4k\hat{k}$ Hence, length of median $= |\vec{AP}| = \sqrt{16+1+16} = \sqrt{33}$

Benchmark Problem 3: KCET 2024 (Q#4)

The volume of the parallelepiped whose co terminous edges are $j\hat{i}+k\hat{j}$, $i\hat{j}+k\hat{k}$ and $i\hat{j}+j\hat{k}$ is

- (A) 6 cu units
- (B) 2 cu units
- (C) 4 cu units
- (D) 3 cu units

Correct Answer: None

Solution Analysis: The volume of the parallelepiped whose coterminous edges are $j\hat{i}+k\hat{j}$, $i\hat{j}+k\hat{k}$ and $i\hat{j}+j\hat{k}$ $= \begin{vmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{vmatrix} = 0-1(0-1)+1(1-0) = -1(-1)+1 = 2\text{cu units}$

Benchmark Problem 4: KCET 2023 (Q#4)

If $a+2b+3c=0$ and $(a \times b)+(b \times c)+(c \times a)=\lambda(b \times c)$, then the value of λ is equal to

- (A) 3
- (B) 4
- (C) 6
- (D) 2

Correct Answer: None

Solution Analysis: Given, $a+2b+3c=0$ $(a+2b+3c) \times c=0 \Rightarrow c \times a=2(b \times c)$ $(a+2b+3c) \times b=0 \Rightarrow a \times b=3(b \times c)$ So, $(b \times c)+(c \times a)+(a \times b)=\lambda(b \times c)$ $(b \times c)+2(b \times c)+3(b \times c)=\lambda(b \times c)$ $6(b \times c)=\lambda(b \times c)$ On comparing with coefficient of $(b \times c)$, we get $\lambda=6$

Benchmark Problem 5: KCET 2020 (Q#1)

If the vectors $2i\hat{j}+3j\hat{k}+4k\hat{i}$, $2i\hat{j}+j\hat{k}-k\hat{i}$ and $\lambda i\hat{j}-j\hat{k}+2k\hat{i}$ are coplanar, then the value of λ is

- (A) 6
- (B) - 5
- (C) - 6
- (D) 5

Correct Answer: None

Solution Analysis: Given vectors $2i\hat{j}-3j\hat{k}+4k\hat{i}$, $2i\hat{j}+j\hat{k}-k\hat{i}$ and $\lambda i\hat{j}-j\hat{k}+2k\hat{i}$ are coplanar $\therefore \begin{vmatrix} 2 & -3 & 4 \\ 2 & 1 & -1 \\ \lambda & -1 & 2 \end{vmatrix} = 0$
 $\Rightarrow 2(2-1)+3(4+\lambda)+4(-2-\lambda)=0 \Rightarrow 2+12+3\lambda-8+4\lambda=0 \Rightarrow 6-\lambda=0 \Rightarrow \lambda=6$